Wildcat Tank Part 2 Official Manual

Pillars of Tech and Altamont Creek Elementary present…

The Wildcat Tank, Part 2!

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1. What is the Wildcat Tank?

The Wildcat Tank is a fun showcase where students use **STEM** to invent a product that improves everyday life. Each student or group presents their idea with a **presentation board** that shows drawings of the product, its purpose, and its key features. You don't have to build a model. **The idea and the pitch are what matter most!**

On event day, you'll pitch your idea to a panel of judges with real STEM careers (scientists and engineers from places like Lawrence Livermore National Laboratory and Quest Science Center). Then you'll answer a few quick questions.

Back for Part 2: last spring, 20 young inventors pitched ideas like sign-language tech, smart water bottles and plant helpers. See their pitches and scores at pillarsoftech.org/wildcat-tank.

Prizes

- **1st Place for Elementary:** $75 gift card
- **1st Place for Middle School:** $75 gift card
- **The Golden Gatty Award:** voted by **THE** Ms. Gatty, prize TBD
- **Every presenter** gets a Wildcat Tank Inventor certificate

Our goal is to give students a hands-on STEM learning experience and bring the community together within the LVJUSD school district. Free food, drinks and STEM activities for the whole family!

2. Who Can Participate and Why Should You?

All TK–5th grade students in the LVJUSD school district can enter. You can enter alone or in a group of up to 3. All levels are welcome. You don't need any experience.

Why join? You get to invent something, practice speaking in front of real scientists, and maybe win a prize. Plus, it makes a great Thanksgiving-break project!

Spots are limited to 28 presenters. After that, you'll go on the waitlist.

3. Key Dates

Date	What happens
Now – Sun, Nov 29	Register at pillarsoftech.org/wildcattankwinter
Nov 23–27 (Thanksgiving break)	Build your board!
Sat, Nov 28	We email you your **slot number and time**
Sun, Nov 29	Last day to register. Slides are due if you're using them (optional)
Mon, Nov 30, 8 PM	**Reply YES to confirm your spot.** If you don't confirm, your slot goes to the waitlist
Wed, Dec 2, 5:30–8:00 PM	Wildcat Tank! Check-in 5:30–5:50 at Altamont Creek Elementary MPR

4. Rules + Guidelines

When you make your board and pitch, you need to include everything on this checklist **at minimum**. You can always add more!

- □ Product name (a slogan is optional)
- □ Drawings: front, back and side views, with labels
- □ Product purpose: what problem it solves and who it helps
- □ Features: 3–6 key features
- □ STEM connection: how it uses STEM
- □ Short pitch: memorized or on note cards

Optional: isometric (3D) drawing, prototype or model, slides, cost estimate.

The pitch is 2 minutes max. The judges then get up to 1 minute for questions. A volunteer will show you time cards at 1:00, 1:30 and STOP.

Mandatory:

- Must be school-appropriate and safe
- Original ideas are encouraged (improvements to existing products are okay)
- No working electronics required (concepts are fine)
- No food, liquids, chemicals or anything that plugs into the wall at your table may be brought to presentations.
- Do not bring drones, planes, or anything that is able to fly.
- Respectful behavior and sportsmanship

5. Judging Rubric

Each judge scores your pitch from **1–10** in five categories, for a total of 50 points per judge.

Category	What judges look for
Problem + usefulness	Is it a real problem? Would people actually use this?
Creativity + innovation	Is it new, or a clever twist on something that exists?
STEM connection	Does the STEM make sense? Did you think it through?
Design + drawings	Are the drawings clear and labeled?
Pitch + confidence	Were you clear, confident and good at answering questions?

We'll have two presentation rooms. The **top 2 from each room** come back for a short final round in front of all the judges, and that round decides 1st, 2nd and 3rd place.

6. Event Day: Your Time Slot

Please read this part carefully. It keeps the night running on time for everyone!

- **You'll get a slot number and a time** by email on Sat, Nov 28 (for example: *Slot 4, about 6:25 PM*). We'll text a reminder on the afternoon of Dec 2.
- **Check-in is 5:30–5:50 PM. Everyone checks in by 5:50, even if your slot is later.** You'll get a sticker with your number. Then set up your board at the gallery tables.
- **Arriving late?** You can still present, but you'll **move to the end of order**. We can't hold the line for anyone.
- **When it's almost your turn,** a volunteer will call you to the **on-deck chairs** next to the stage. Please stay near your board until you've presented.
- **Plan to stay until about 8:00 PM.** Finals and awards are at the end, and every presenter gets a certificate on stage!
- **Can't make it after all?** Email us as soon as you know so we can give your slot to someone on the waitlist.

Night at a glance:

Time	What's happening
5:30–5:50	Check-in and set up your board
5:50	Quick presenter meeting (everyone!)
6:00	Welcome + meet the judges
6:10–7:20	Presentations, Families: gallery walk, STEM station
7:20–7:30	Food break, STEM Station
7:30–7:45	Finals
7:45–8:00	Awards + certificates for everyone

7. Brainstorming Tips

You don't need a “crazy” idea. The best inventions solve small, everyday problems.

Think about:

- Things that waste time at school or at home
- Safety issues, or things you lose a lot
- Messy or annoying problems you wish were easier

Quick question: “What do I wish existed?”

NEW — Ask your family: Interview one family member: “What's the most annoying thing you do every day?” Their answer might be your invention!

8. Quick Brainstorm Methods

- **Annoyance list:** Write down 10 everyday annoyances, then pick 1 to solve.
- **Combine two things:** Mix two objects together (for example, a backpack + a charger).
- **SCAMPER:** Improve something that already exists: **S**ubstitute, **C**ombine, **A**dapt, **M**odify, **P**ut to another use, **R**everse/**R**earrange.
- **5 Whys:** Keep asking “why?” until you find the real problem.

9. STEM Connection Tips

STEM doesn't have to mean a robot. Smart design counts too.

STEM can be:

- **Science/materials:** waterproof, reusable, insulated
- **Technology:** sensors, alarms, apps
- **Engineering:** moving parts, structure, balance
- **Math/data:** measuring, tracking, timing

Sentence starter: “My product uses STEM because it uses ___ to ___.”

10. Drawing Tips

Required: front, back and side views.

Make it stronger by adding:

- Labels on the parts
- Arrows showing what moves or opens
- Callout bubbles that explain features
- *Bonus:* isometric (3D) view, dimensions, or an “in use” drawing showing someone using it

Tip: Draw in pencil first, then trace in marker. Make your drawings big enough for the judges to see from 6 feet away.

11. Pitch Tips

Keep it simple: **problem → solution → why it matters.**

Easy structure (fits in 2 minutes):

1. Hook (“Have you ever…?”): about 10 sec
2. The problem: about 15 sec
3. Your product and what it does: about 20 sec
4. How it works (1–2 steps): about 20 sec
5. Best features: about 30 sec
6. Closing line: about 15 sec

NEW — Time yourself! Practice with a timer. If you're over 2 minutes, cut a feature, not your hook.

12. Public Speaking Tips

- Speak slower than you think you need to
- Make eye contact and use a clear voice
- Practice out loud 2–3 times (try it for your family at Thanksgiving dinner!)
- If you mess up, pause and keep going
- Skip filler words like “uhhh,” “you know” and “so basically.” When you get stuck, a short pause is often the best fix.

13. Judge Q&A Tips

Answer clearly first, then add details. It's okay to say “I haven't figured that out yet, but I would…”

Questions judges often ask:

- Who is it for?
- How does it work?
- What materials would you use? How much would it cost?
- What makes it different from what's already out there?
- What challenges might come up?

14. Presentation Display Tips

A trifold board is the standard way to present, and it's all you need. It goes on your gallery table, and you'll bring it with you when you pitch.

Using slides (optional):

- Send your Google Slides or Canva link, or your PowerPoint or Keynote file, to pillarsoftech@gmail.com by **Sun, Nov 29**.
- We'll load all slides on our laptop in pitch order. **Personal laptops can't be plugged in on event night.** This keeps every slot on time.
- Want to show a video or app demo? Tell us by Nov 29 so we can test it ahead of time.

A strong board includes:

- Product name, big, at the top
- Problem + who it helps
- Drawings (front, back, side), labeled
- 3–6 features
- “How it uses STEM”
- Your name(s), grade and school on the back

15. What to Bring Checklist

- ☐ Your presentation board/material
- ☐ Note cards (if you're using them)
- ☐ Prototype or model (optional)
- ☐ Your slot number email (on a phone is fine)
- ☐ Your family!

16. Parent FAQ

Does it cost anything? No. Entry, food and activities are all free.

Does my child need to build something? No. A drawing and a pitch are enough.

Can siblings enter together? Yes, as a group of up to 3. Siblings entering separately will get back-to-back slots.

How much help can parents give? Help with supplies and practice is great. The idea and the words should be your child's own. Judges can usually tell!

Can younger siblings come? Yes! There are STEM stations for all ages while pitches happen.

Will there be photos? Pillars of Tech may take photos and video for our website and social media. Tell us at check-in if you'd prefer your child not be pictured.

Questions? Email pillarsoftech@gmail.com. Event info: pillarsoftech.org/wildcattankwinter https://luma.com/us8xb8om · Instagram: @thepillarsoftech